Supplementary Material: Design Equations and Parametric Sizing for Hierarchical Coordination in Large Autonomous Space Swarms

Thijs Hakkenberg Project Dyson, Open Research Initiative
Email: thijs@hakkenberg.com
https://projectdyson.org

***Abstract*—This document provides supplementary material for "Design Equations and Parametric Sizing for Hierarchical Coordination in Large Autonomous Space Swarms." It contains detailed derivations, extended tables, worked examples, and placeholder sections for NS-3 validation results that accompany the core paper. Section references of the form "§X (core)" refer to the corresponding section of the main paper.**

APPENDIX

This section provides the complete notation reference for all symbols used in the core paper and supplement, extending the compressed notation table in §I (core). Symbols are organized by functional group.

This section provides the full γ decomposition and per-slot timing ledger summarized in §IV-J and §V-C of the core paper. All γ values are computed from the time-domain form (Eq. 3 of the core paper) using unrounded intermediate values.

A. γ Decomposition Table

Table S2 decomposes the TDMA slot into its constituent components under CCSDS Proximity-1 framing at 24 kbps PHY rate. The same structure applies at other PHY rates; only the time column rescales (fixed-time overheads T_{guard} and T_{acq} remain constant, compressing γ at higher rates).

B. Slot Timing Ledger

Table S3 is the authoritative timing ledger partitioning every per-slot overhead as either "in γ" (absorbed by the slot efficiency equation) or "outside γ" (separately budgeted in Test B margin). This ledger is the single reference for all feasibility claims.

C. Measurement Protocol for γ

To anchor γ in hardware measurements: (a) measure T_{acq} (P95) across representative SNR and Doppler conditions; (b) measure turnaround time and timing jitter to determine T_{guard}; (c) compute γ via Eq. 3 (core); (d) propagate uncertainty using the closed-form sensitivity: $\partial R_{\text{PHY,min}}/\partial T_{\text{acq}} = (k_c-1)/(T_c\gamma)$. At nominal parameters this gives ≈ 13.3 kbps per second of T_{acq} change.

D. γ Consistency Ledger

The authoritative γ values (all from Eq. 3 of the core paper with unrounded intermediates) are: 24 kbps: $T_{\text{slot}}=112.2$ ms, $\gamma_{24}=0.761$; 30 kbps: $T_{\text{slot}}=91.7$ ms, $\gamma_{30}=0.745$; 35 kbps: $T_{\text{slot}}=80.0$ ms, $\gamma_{35}=0.732$; 50 kbps: $T_{\text{slot}}=59.0$ ms, $\gamma_{50}=0.695$. All Model C cold-start ($T_{\text{acq}}=5$ ms, $T_{\text{guard}}=4.7$ ms). ACK (0.5 ms per node) is *not* included in γ; it is separately accounted in Test B margin. Figs. 4, 6 and Tables 4, 5, 6, 7 of the core paper use these values. Model S ($\gamma_{24}=0.949$) appears only in the joint interaction table (illustrative; not for design).

This section provides the complete GE sensitivity analysis and inter-cycle recovery derivation summarized in §IV-C of the core paper. The GE model is a *what-if design tool*; all results are conditional on unmeasured parameters.

E. Inter-Cycle Recovery Derivation

Under the GE model with per-cycle state transitions ($p_{GB}=0.05$, $p_{BG}=0.50$), each node's ISL experiences independent GE state transitions. The per-cycle recovery probability after k cycles is:

$$p_{\text{rec}}^{(k)} = \pi_G^{(k)}(1-p_G^{M_r+1}) + \pi_B^{(k)}(1-p_B^{M_r+1}) \tag{1}$$

where $\pi_G^{(k)}$ and $\pi_B^{(k)}$ are the state probabilities at cycle k, evolving via the Markov transition matrix:

$$\begin{bmatrix}\pi_G^{(k+1)}\\ \pi_B^{(k+1)}\end{bmatrix} = \begin{bmatrix}1-p_{GB} & p_{BG}\\ p_{GB} & 1-p_{BG}\end{bmatrix}\begin{bmatrix}\pi_G^{(k)}\\ \pi_B^{(k)}\end{bmatrix} \tag{2}$$

The cumulative recovery CDF is:

$$F(k) = 1 - \prod_{j=1}^{k}(1-p_{\text{rec}}^{(j)}) \tag{3}$$

At default parameters ($p_{BG}=0.50$, $p_B=0.90$, $M_r=2$): mean recovery = 1.7 cycles, P95 = 4 cycles (DES-verified with 30 MC replications, $N=10{,}000$).

F. Coherence Regime Discussion

The GE model transitions once per T_c; intra-cycle retransmissions face the same channel state. This makes intra-cycle

TABLE S1
COMPLETE NOTATION REFERENCE

Symbol	Definition	Default	Group
Fleet and Topology			
N	Fleet size (nodes)	10^3–10^5	Fleet
k_c	Cluster size (nodes per coordinator)	100	Fleet
k_r	Clusters per regional coordinator	$\lceil N/(k_c \cdot n_r)\rceil$	Fleet
n_r	Number of regional coordinators	10	Fleet
f_{RF}	Fraction of fleet in RF-backup mode	0.012	Fleet
F	Number of orthogonal RF channels	8	Fleet
R	Spatial reuse factor	7	Fleet
Cluster Coordination			
T_c	Coordination cycle period	10 s	Cluster
C_{node}	Per-node bandwidth allocation	1 kbps	Cluster
S_{eph}	Status report size	256 B	Cluster
S_{cmd}	Command message size	512 B	Cluster
C_{coord}	Coordinator ingress link rate (kbps)	—	Cluster
$C_{\text{coord,info}}$	Coordinator information-rate requirement	20.2 kbps	Cluster
α_{RX}	Ingress fraction of T_c ($= T_{\text{ing}}/T_c$)	0.908 (30 kbps)	Cluster
d	Campaign duty factor	$[0, 1]$	Cluster
p_{cmd}	Per-cycle command probability (given ON)	$[0, 1]$	Cluster
q	Unicast fraction of commands	$[0, 1]$	Cluster
L_{cmd}	Command stagger cycles	19 (30 kbps)	Cluster
p_{exc}	Per-node per-cycle reporting probability	0.10–1.0	Cluster
f_{decision}	Raft decisions per cycle	1	Cluster
N_R	Raft protocol rounds per decision	2	Cluster
TDMA Frame			
$\gamma(R_{\text{PHY}})$	TDMA slot efficiency	0.732–0.761	TDMA
R_{PHY}	Physical-layer burst rate	24–50 kbps	TDMA
R_{FEC}	FEC code rate (LDPC)	7/8	TDMA
O_{frame}	Framing overhead (ASM+addr+ctrl+FCS)	104 bits	TDMA
T_{payload}	Information-bit transmission time ($S \times 8/R_{\text{PHY}}$)	rate-dependent	TDMA
T_{slot}	Total slot duration	80–112 ms	TDMA
T_{guard}	Guard interval (prop+turn+jitter)	4.7 ms	TDMA
T_{acq}	Carrier acquisition time (cold-start)	5 ms	TDMA
Channel Model			
p_{BG}, p_{GB}	GE state transition probabilities	0.50, 0.05	Channel
p_G, p_B	Per-packet loss probability in Good/Bad state	0.01, 0.90	Channel
M_r	Maximum retransmission attempts per cycle	0–2	Channel
π_G, π_B	GE stationary state probabilities	0.909, 0.091	Channel
Overhead Metrics			
η	Protocol overhead ($= \eta_0 + d \cdot \eta_{\text{cmd}}$)	5.6–46%	Overhead
η_0	Architecture-specific overhead	5.6%	Overhead
η_{cmd}	Workload-dependent overhead (at $p_{\text{cmd}} = 1$)	41%	Overhead
η_{total}	Total utilization ($= \eta + 20.5\%$ baseline)	26–67%	Overhead

TABLE S2
γ DECOMPOSITION FROM CCSDS PROXIMITY-1 FRAMING
($S_{\text{EPH}} = 256$ B, 24 KBPS PHY)

Component	Bits	Time (ms)	Source
Payload ($S \times 8$)	2,048	85.33	—
Framing (ASM+addr+ctrl+FCS)	104	4.33	CCSDS 211.0-B-5
FEC parity (LDPC 7/8)‡	308	12.83	CCSDS 131.0-B-4
Guard (prop+turn+jitter)	—	4.70	Prox-1 + geometry
Acquisition (pointing dwell)	—	5.00	Assumed; Prox-1 class
Total slot	**2,460**	**112.20**	
$\gamma_{C,24}$	$85.33/112.20 = \mathbf{0.761}$		Eq. 3 (core)
Model S (no FEC/acq)	$85.33/90.03 = 0.949$		Eq. 2 (core)

‡Rate 7/8 LDPC selected to maximize throughput given sufficient ISL link margin (>6 dB at 30 kbps); rate-1/2 doubles symbol time, making 30 kbps infeasible. All γ values computed from Eq. 3 (core) with unrounded intermediates.

ARQ structurally ineffective when the coherence time $\tau_c \geq T_c$: under bad-state bursts, all $M_r + 1$ attempts face $p_B = 0.90$, yielding only 27.1% recovery with $M_r = 2$.

This conclusion is *conditional* on slow fading ($\tau_c \geq T_c$). For fast-fading ($\tau_c \ll T_c$, e.g., rapid tumbling, multipath from reflective surfaces): channel state decorrelates between retransmissions and delivery reaches 98.9% with $M_r = 2$—intra-cycle ARQ is fully effective.

Physical mapping: $p_{BG} \approx T_c/\bar{T}_B$; default 0.50 is illustrative (no ISL measurements exist). LEO mechanisms include: structural shadowing ($p_{BG} \geq 0.50$), mispointing ($p_{BG} \sim 0.10$), and occultation ($\sim$35 min; use schedules). Geometric justification: 1 m panel at 2 m separation, 2°/s tumble rate gives $\tau_c \approx 14$ s $> T_c$; stabilized spacecraft: 50–300 s $\gg T_c$.

G. ARQ Viability Analysis

ARQ provisioning allocates M_r shared retransmission slots within the superframe margin; the available window is $T_c - T_{\text{ing}} - T_{\text{egr}} - T_{\text{ACK}}$: 680 ms at 30 kbps, 1,830 ms at 35 kbps.

TABLE S3
SLOT TIMING LEDGER: IN-γ VS. OUTSIDE-γ COMPONENTS (MODEL C, 30 KBPS)

Component	Bits	ms	Scope
In γ (per slot)			
Payload ($S \times 8$)	2,048	68.27	Data
Framing (ASM+addr+ctrl+FCS)[a]	104	3.97	CCSDS 211.0-B-5
FEC parity (LDPC 7/8)	308	10.27	CCSDS 131.0-B-4
Guard (prop+turn+jitter)	—	4.70	Geometry
Acquisition (cold-start)	—	5.00	Assumed
Outside γ (per cycle)			
ACK mini-slot ($k_c \times 0.5$ ms)	—	50	Conservative
TX/RX turnaround (2×2 ms)	—	4	Half-duplex
Re-sync preamble (120 bits)	120	4	Post-turnaround
Ranging + sync + drift	—	19	CCSDS 414.0-G-2; TC
In-γ subtotal (per slot)	**2,460**	**92.2**	$\gamma_{30} = 0.745$
Outside-γ subtotal (per cycle)	—	**77**	Test B margin

[a]Framing bits FEC-encoded per CCSDS practice ($O_{\text{frame}}/(R_{\text{FEC}}R_{\text{PHY}})$). Authoritative γ from Eq. 3 (core) with unrounded intermediates.

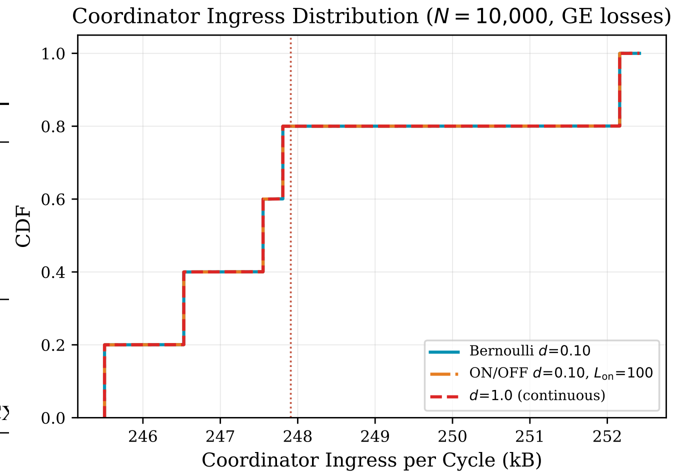

Fig. S1. CDF of per-cycle coordinator ingress bytes ($N = 10{,}000$, $k_c = 100$, GE). Bernoulli $d = 0.10$ (solid): bimodal. ON/OFF $d = 0.10$, $L_{\text{on}} = 100$ (dash-dot): heavier P99. Continuous $d = 1.0$ (dashed): full load. Vertical lines: closed-form means. Buffer factor $M = 1.30$ keeps overflow $< 1\%$ at $d = 0.10$.

Expected demand: Under GE with $\pi_B = p_{GB}/(p_{BG} + p_{GB}) = 0.091$, $E[\text{failed}] = k_c \pi_B p_B = 100 \times 0.091 \times 0.90 \approx 8.2$ nodes; P95 (binomial, $n = 100$, $p = 0.082$) ≈ 14 nodes $\times 92$ ms $= 1{,}288$ ms. At 30 kbps: 680 ms $< 1{,}288$ (P95 infeasible); at 35 kbps: 1,830 ms $> 1{,}288$ (P95 feasible).

Coherence-regime sensitivity (what-if; no ISL measurements): (i) $\tau_c \geq T_c$ *and* $p_B \geq 0.7$: use 35 kbps + inter-cycle recovery (P95 $= 4$ cycles at $p_{BG} = 0.50$); (ii) $\tau_c \ll T_c$ or $p_B < 0.7$ (benign channel): ARQ effective (98.9%), 30 kbps suffices. Select R_{PHY} from Fig. 6(b) of the core paper using measured τ_c/T_c and p_B.

M_r **selection trade:** $M_r = 0$: no intra-cycle ARQ; accept inter-cycle recovery (AoI penalty $= +T_c$ per lost cycle; P95 $= 4$ cycles); 30 kbps sufficient, full 680 ms margin preserved. $M_r = 1$: recover $\sim 27\%$ of slow-fading losses intra-cycle; requires 35 kbps (680 ms margin insufficient at 30 kbps). $M_r = 2$: marginally improves recovery to $\sim 35\%$ under slow fading but consumes ~ 184 ms additional margin; diminishing returns unless $\tau_c \ll T_c$.

H. Sensitivity Sweep

Fig. 6(b) of the core paper presents the primary sensitivity result: P95 recovery vs. p_{BG} for $p_B \in \{0.70, 0.90, 0.95\}$. At $p_{BG} = 0.50$: P95 = 3–5 cycles; at $p_{BG} = 0.10$ (persistent bad state): 12–18 cycles. Missions should use these design curves with measured τ_c.

This section provides the full distributional analysis and DES architectural details summarized in §III-A and §IV-F of the core paper.

I. V&V Tiers

The verification and validation hierarchy follows IEEE 1012: **Tier 1:** DES reproduces analytical means ($<$0.1% deviation); this confirms implementation correctness, not model validity. Distributional tails (Fig. S1) are the DES's incremental contribution beyond closed-form analysis. **Tier 2:** slot-level simulation reveals ARQ$\times$TDMA coupling; packet-level simulation anchors γ (parameter anchoring, not independent validation). **Tier 3 (absent):** ISL measurements, NS-3 MAC simulation, hardware γ measurement (§X). No external validation exists.

J. Distributional Tail Analysis

The DES's sole incremental contribution beyond mean-value agreement is distributional tails under campaign burstiness. Under ON/OFF Markov burst models ($L_{\text{on}} = 100$ cycles), the P99 coordinator ingress shifts $\sim 25\%$ above the Bernoulli (independent duty-cycle) baseline. A buffer factor $M = 1.30$ at $d = 0.10$ ensures $<$1% overflow; $M = 1.50$ is required for $d = 0.50$.

These are conditional predictions under the Markov ON/OFF assumption, not validated findings. Under heavy-tailed ON durations (e.g., Pareto-distributed campaign lengths), M would increase; under lighter tails, M would decrease.

K. MMPP/D/1 Corroboration

Under compound campaign + GE dynamics, the arrival process at the coordinator is well-approximated as MMPP/D/1 (Markov-Modulated Poisson Process with deterministic service). An MMPP/D/1 fluid bound corroborates $M \approx 1.25$–1.35 for the assumed parameters. P99 buffer occupancy is $\sim$20–30% above deterministic; closed-form bounds for the exact MMPP/D/1 tail are future work.

This section provides the full thundering-herd analysis for coordinator election under RF-backup conditions, extending the summary in §III-B of the core paper.

L. Slotted ALOHA / BEB Analysis

When a coordinator fails and optical ISL is unavailable, $k_c = 100$ nodes simultaneously attempt Raft election via UHF

(2.5 kbps). Under Slotted ALOHA the initial offered load is $G = 25$ (collapse regime; throughput $S = Ge^{-G} \approx 0$).

Binary Exponential Backoff (BEB) progressively reduces contention:

- Round 0: $W_0 = 4$, $G_0 = 100/4 = 25$ (collapse)
- Round 1: $W_1 = 8$, $G_1 = 100/8 = 12.5$ (still overloaded)
- Round 2: $W_2 = 16$, $G_2 = 100/16 = 6.25$ (overloaded)
- Round 3: $W_3 = 32$, $G_3 = 100/32 = 3.13$ (marginal)
- Round 4: $W_4 = 64$, $G_4 = 100/64 = 1.56$ (stable: $S \approx$ 825 bps)

At $G_4 = 1.56$, throughput $S = Ge^{-G} \times R_{\text{UHF}} = 1.56 \times e^{-1.56} \times 2{,}500 \approx 825$ bps. Total election traffic (Raft): 12.8 KB = 102,400 bits. Expected election time: 102,400/825 $\approx$ 124 s plus BEB convergence overhead, yielding $\sim$140–160 s total. Raft's randomized timeout (150–300 ms) partially mitigates by staggering candidates.

M. Election–Status Interaction

During the $\sim$160 s RF-backup election, status reporting is *suspended* (hierarchical coordination suspended; see below). Election traffic thus has exclusive use of the 2.5 kbps UHF channel. AoI impact of the suspension: +16 missed cycles (+160 s), which is modest relative to AoI P99 = 441 s.

If status reporting were *not* suspended, the combined load (election 640 bps + 100 beacons $\times$ 51 bps = 5,740 bps) would exceed UHF capacity (2.5 kbps), causing election failure. Suspension is therefore a *design requirement*, not an optimization.

N. Coordination Suspension Design

Coordinator ingress ($\geq$27 kbps) exceeds RF-backup capacity (2.5 kbps) by $>$10$\times$; hierarchical coordination is *suspended* during RF-backup operation (beacon-only mode, $\eta \approx 5.7\%$). During suspension, nodes rely on passive safety: conjunction probability per node per 300 s $\approx 10^{-8}$; autonomous collision avoidance is not required during the $\sim$160 s election window. TDMA sizing (Sections IV–V of the core paper) applies to S-band only.

Gaps between elections: $<$ 1/yr per cluster (based on 2%/yr node failure rate applied to coordinators). The compound probability of a coordinator failure occurring during a GE bad-state (triple fault) is 1.8×10^{-5}/yr, extending election to $\sim$310 s.

This section provides the full dynamic topology analysis under J2 perturbations, extending the summary in §V-D of the core paper.

O. Walker Constellation Analysis

J2 analysis (Vallado, *Fundamentals of Astrodynamics*) for a Starlink-like Walker constellation (53° inclination, 550 km altitude, 72 planes $\times$ 22 satellites/plane):

- **Co-orbital re-association:** zero (nodes in the same orbital plane experience identical J2 precession)
- **Cross-plane re-association:** $\sim$0.014/orbit fleet-wide (14.8 kB state transfer, <0.3% of per-node budget per cycle)
- **AoI transient:** 33 s per re-association event (3–5 s handoff + state rebuild)

Heterogeneous orbits (mixed altitudes or inclinations) produce higher differential precession rates; the re-association frequency scales as $\propto \Delta\Omega/T_c$ where $\Delta\Omega$ is the differential RAAN precession rate between adjacent planes.

P. Worst-Case Boundary Cluster

At orbital plane interfaces, re-association occurs once per orbit ($\sim$90 min) as nodes drift between clusters. At 3–5 s optical handoff per event, duty loss $\approx$ 0.06–0.09% of cycle time—negligible for TDMA margin but relevant for coordinator state continuity. Mitigation: overlap zones with dual-cluster membership during handoff; state pre-staging via optical ISL.

DES limitation: static cluster membership is assumed in all DES runs. Dynamic re-association adds <0.5% overhead based on the analytical estimates above; DES verification of dynamic membership is future work.

This section provides the extended topology comparison summarized in §IV-G of the core paper, including the sectorized mesh model and intermediate architecture bounds.

Q. Sectorized Mesh Model

Global-state mesh (full fleet state per node): $O(N^2)$ messages/cycle via gossip; at $N = 10^5$, per-node traffic $\approx$ 73 MB/cycle ($\gg$ 1 kbps budget). The sectorized mesh reduces this by limiting each node to $k_s = \lceil\sqrt{N}\rceil$ neighbors with a fanout cap of 10: at $N = 10^5$, $k_s = 317$, capped at 10 neighbors. Stress-case overhead: $\eta \approx$ 65–67% with only 3.2% fleet state coverage.

Both global-state and sectorized mesh are intentional *upper bounds* demonstrating that flat topologies are infeasible at scale under the 1 kbps per-node budget. Hierarchical aggregation achieves $O(k_c)$ per-node cost via coordinator summarization.

R. Centralized Bounds

Centralized ground processing: $M/D/1$ queue model; $N_{\text{max}} = c\mu_s/r$ (10^4 at $c = 1$ server, 10^6 at $c = 100$). This is an intentional compute bound—propagation latency and uplink spectrum constraints bind first in practice. The centralized model provides a lower bound on protocol overhead (no intra-fleet coordination traffic) but with single-point-of-failure and latency constraints.

Centralized with regional aggregation retains the single-point and latency constraints of pure centralized operation while adding one aggregation layer. This intermediate architecture is not analyzed in detail as it does not change the scaling class ($O(N)$).

S. Partial-State Mesh

k-hop gossip ($O(kN)$ messages/cycle) provides partial fleet state without global guarantees. At $k = 3$: $\sim$30% of fleet state reachable per node; overhead $\eta \approx$ 15–20% (stress). The partial-state mesh trades coverage completeness for reduced overhead but cannot guarantee coordination consistency (split-brain risk under partitions). The centralized and full-mesh bounds bracket the viable design space; hierarchy occupies the favorable region.

This section provides the full link budget analysis supporting the PHY rate selection in §IV-A of the core paper.

T. S-Band ISL Parameters

S-band 2.2 GHz inter-satellite link budget:

- Transmit power: 1 W (+30 dBm)
- Antenna gain: 6 dBi patch (both ends)
- EIRP: +36 dBm
- Range: 500 km (intra-cluster nominal)
- Free-space path loss: −151.2 dB (at 2.2 GHz, 500 km)
- System noise temperature: 500 K (T_{sys}; includes antenna, LNA)
- $C/N_0 = 64.5$ dB-Hz
- At 24 kbps: $E_b/N_0 = 64.5 - 10\log_{10}(24{,}000) = 20.7$ dB; margin $= 20.7 - 4.8 = 15.9$ dB (LDPC 7/8 threshold)
- At 35 kbps: $E_b/N_0 = 64.5 - 10\log_{10}(35{,}000) = 19.1$ dB; margin $= 19.1 - 4.8 = 14.3$ dB
- Aggregate capacity: >200 kbps at 6 dB margin threshold
- Per-node allocation: 1 kbps at $k_c = 100$ with ∼50% margin

U. UHF RF-Backup Parameters

UHF omnidirectional backup link:

- Frequency: 400 MHz band
- Transmit power: 0.5 W (+27 dBm)
- Antenna gain: 0 dBi (omnidirectional)
- $E_b/N_0 = 16.5$ dB at 1 kbps
- Maximum sustainable rate: ∼2.5 kbps
- Purpose: safe-mode beacons and coordinator election only

RF backup is necessary because tumbling spacecraft cannot maintain optical pointing. The hierarchy is suspended during RF-backup operation (§K).

V. Sensitivity Analysis

± 3 dB link margin variation maps to per-node budget $\in [0.5, 2]$ kbps:

- At 0.5 kbps (−3 dB): stress-case $\eta_{\text{total}} \approx 92\%$ (infeasible; TDMA margin negative)
- At 1 kbps (nominal): $\eta_{\text{total}} \approx 67\%$ (feasible; TDMA margin 680 ms at 30 kbps)
- At 2 kbps (+3 dB): $\eta_{\text{total}} \approx 33\%$ (comfortable; TDMA non-binding)

At ≥2 kbps, stress-case η halves to $\sim 23\%$ and TDMA sizing becomes non-binding; only Test A (byte budget) is required.

This section provides the full γ-conditional lookup table and acquisition mode sensitivity analysis from §IV-J of the core paper, intended as a practitioner reference.

TABLE S4
γ-CONDITIONAL PHY RATE LOOKUP ($k_c = 100$, $S = 256$ B, $T_c = 10$ s)

Measured γ	$R_{\text{PHY,min}}$	**Margin**	**Application Note**
$[0.50, 0.65)$	≥50 kbps	>40%	Conservative; high-overhead modem
$[0.65, 0.70)$	40 kbps	20–25%	Includes ARQ margin
$[0.70, 0.80)$	**35 kbps**	**12–19%**	**Recommended (CCSDS default)**
$[0.80, 0.90)$	30 kbps	7–15%	Minimum; no ARQ margin
$[0.90, 1.00)$	25 kbps	5–10%	Ideal modem (continuous tracking)

Margin $= (T_c - T_{\text{ing}} - T_{\text{egr}} - T_{\text{ACK}})/T_c$ (Table S3). Practitioners: measure γ per the protocol in §A, then select R_{PHY} from this table. For $k_c \neq 100$ or $S \neq 256$ B: recompute via Algorithm 1 (core).

TABLE S5
γ CONSISTENCY LEDGER (MODEL C, COLD-START)

R_{PHY} (kbps)	T_{payload} (ms)	T_{slot} (ms)	γ	Feasibility
24	85.33	112.2	0.761	Infeasible (−1,300 ms)
30	68.27	91.7	0.745	Min. viable (680 ms)
35	**58.51**	**80.0**	**0.732**	**Recommended (1,830 ms)**
50	40.96	59.0	0.695	Ample (>4,000 ms)

W. Full Consistency Ledger

The γ consistency ledger ensures all tables, figures, and equations in both the core paper and supplement use identical values:

X. Acquisition Mode Sensitivity

Three acquisition modes are evaluated: *cold-start* ($T_{\text{acq}} =$ 5 ms, conservative, CCSDS Prox-1 default), *reacquire* ($T_{\text{acq}} =$ 2 ms, carrier relock after brief interruption), *tracking* ($T_{\text{acq}} = 0$, continuous carrier lock).

At cold-start (including ACK): 30 kbps margin = 680 ms (6.8% of T_c); 35 kbps margin = 1,830 ms (18.3%); 24 kbps deficit = −1,300 ms (infeasible). At reacquire: 30 kbps margin = 1,068 ms (10.7%).

Breaking points: 35 kbps remains feasible up to $T_{\text{acq}} \approx$ 19 ms; 30 kbps up to $T_{\text{acq}} \approx$ 7.4 ms. **Slow-acquisition hardware** ($T_{\text{acq}} >$ 10 ms, common in low-cost CubeSat radios): recommend $R_{\text{PHY}} \geq 40$ kbps (per Fig. 4 of the core paper). The 35 kbps cold-start recommendation is appropriate for CCSDS-class hardware.

This section provides the full claim-to-evidence mapping from §V-A of the core paper, extended with an NS-3 column for future validation results.

This section describes the planned NS-3 validation methodology for the analytical and DES results presented in the core paper. Results will be populated after NS-3 simulation runs complete.

Y. Simulation Setup

The NS-3 scenario implements a star-topology TDMA cluster using NS-3 3.41+ core networking primitives (`PointToPointNetDevice`, `BurstErrorModel`, `FlowMonitor`). No code or equations are shared with the Python analytical model or DES.

Key differences from the analytical model:

TABLE S6
CLAIM MAP BY EVIDENCE TIER (EXTENDED WITH NS-3 COLUMN)

	Tier 1 (Internal)			Tier 2	Std.-based param.	Tier 3	NS-3
Result	Analyt.	DES	Distrib.	Slot-sim	CCSDS γ‡	Ext.	Sim.§
$\eta_0 \approx 5.6\%$	§IV-E	<0.1%	—	—	—	—	—
$\eta_S \approx 46\%$	§IV-E	<0.1%	CDF	—	—	—	—
Coord. 24–30 kbps	Eq. 6	—	CDF	Exact	24 infeas.	—	—
Margin 680 ms†	Tbl. 4	—	—	Exact	—	—	—
γ sensitivity	Tbl. S5	—	—	Conf.	$\gamma = 0.76$	—	—
ARQ infeas. (GE)	Markov	—	—	52.7%	Conf.	—	—
35 kbps recommended	Tbl. S5	—	—	ARQ	$\gamma_{35} = 0.732$	—	—
1 kbps per-node budget	—	—	—	—	Link budget	—	—
AoI P99 = 440 s	Eq. 5	Conf.	—	—	—	—	—
GE P95 rec. 4 cyc.	Markov	Conf.	CDF	—	—	—	—
$L_{\text{cmd}}(q)$ stagger	Eq. 8	—	—	Conf.	—	—	—
TDMA decoupling	—	—	—	$(M_r = 0)$	—	—	—
ARQ coupling	—	—	—	$(M_r \geq 1)$	—	—	—
T_c^{fleet} reuse $(R = 7)$	Eq. 4	—	—	—	—	NS-3	—
Antenna sched.	—	—	—	—	—	Future	—
MAC contention	—	—	—	—	—	Future	—

†30 kbps Model C (Table 4 of core). Model S at 24 kbps: 611 ms. ‡Packet-level γ is a *standards-based parameter estimate* (§IV-J of core), not independent validation. §NS-3 column: placeholder for future validation runs (§X).

- Framing: 88 bits (ASM+addr+ctrl+FCS) vs. 104 bits in Model C
- Acquisition: stochastic LogNormal($\mu = \ln 5\,\text{ms}$, $\sigma = 0.3$) vs. fixed 5 ms
- Channel: NS-3 `BurstErrorModel` vs. our custom GE implementation
- Scheduling: NS-3 event-driven packet handling vs. time-slot aggregation

Z. Parameter Sweeps

PHY rate: $\{20, 24, 28, 30, 32, 35, 40, 50\}$ kbps. Cluster size: $\{50, 100, 200, 500\}$. GE states: no-loss, default ($p_{BG} = 0.50$, $p_B = 0.90$), severe. ARQ: $M_r \in \{0, 1, 2\}$.

. Results

[Placeholder: full sweep results tables and additional figures.]

This section analyzes four alternative TDMA slot structures beyond the cold-start baseline used in the core paper. Detailed NS-3 comparison results will be added when available.

Four slot structures are analyzed:

Config A (Cold-start, baseline): Single-packet per slot with full carrier acquisition. This is the conservative assumption used throughout the core paper.

Config B (Multi-packet burst): Three packets per slot with a single acquisition phase. Inter-packet gap: 0.5 ms. Amortizes acquisition overhead across 3 payloads.

Config C (Tracking mode): First slot uses cold-start acquisition; subsequent slots maintain carrier tracking ($T_{\text{acq}} = 0$). Models continuous-carrier receivers.

Config D (Bitmap ACK): Same data slots as Config A, but coordinator sends a 13-byte aggregate bitmap ACK instead of individual 0.5 ms ACK mini-slots. Frees coordinator egress time.

[Placeholder: detailed analytical derivations and NS-3 sweep results for each config.]

This section provides the complete sizing walkthrough referenced in §V-C of the core paper, with step-by-step annotations mapping each computation to Algorithm 1 (core).

. Problem Statement

Size a hierarchical coordination cluster with parameters: $k_c = 100$ nodes, campaign duty factor $d = 0.10$, candidate PHY rate 35 kbps. Determine feasibility under both Test A and Test B with $M_r = 1$ intra-cycle retransmission.

. Step 1: Compute γ (Algorithm 1, Line 1)

From the slot efficiency equation (Eq. 3 of the core paper):

$$T_{\text{payload}} = \frac{256 \times 8}{35{,}000} = 58.51 \text{ ms}$$
$$T_{\text{FEC}} = \frac{256 \times 8 \times (1/0.875 - 1)}{35{,}000} = 8.36 \text{ ms}$$
$$T_{\text{framing}} = \frac{104}{0.875 \times 35{,}000} = 3.39 \text{ ms}$$
$$T_{\text{slot}} = 58.51 + 8.36 + 3.39 + 4.70 + 5.00 = 79.97 \text{ ms}$$
$$\gamma_{35} = 58.51/79.97 = 0.732$$

Annotation: The FEC rate 7/8 expands payload by $1/R_{\text{FEC}} - 1 = 1/7$; framing bits (104) are also FEC-encoded per CCSDS practice. Guard (4.70 ms) and acquisition (5.00 ms) are fixed-time overheads independent of PHY rate.

. Step 2: Test A — Byte Budget (Algorithm 1, Line 3)

$$\eta_0 = 5.6\% \quad \text{(heartbeats 5.1\% + summaries 0.4\% + elections 0.1\%)}$$
$$\eta_{\text{cmd}} = d \times p_{\text{cmd}} \times \frac{S_{\text{cmd}} \times 8}{C_{\text{node}} \times T_c}$$
$$= 0.10 \times 1.0 \times \frac{512 \times 8}{1{,}000 \times 10} = 4.1\%$$
$$\eta_{\text{total}} = \eta_0 + \eta_{\text{cmd}} + \eta_{\text{baseline}}$$
$$= 5.6\% + 4.1\% + 20.5\% = 30.2\% \quad \checkmark \ (< 100\%)$$

TABLE S7
SIZING WALKTHROUGH SUMMARY ($k_c = 100$, $d = 0.10$, 35 KBPS)

Step	Test	Result	Status
1	γ_{35}	0.732	—
2	Test A (η_{total})	30.2%	✓
3	Test B (margin)	1,838 ms	✓
4	ARQ ($M_r = 1$)	1,758 ms residual	✓
5	Heuristic	34.8 kbps < 35	✓

Annotation: The 20.5% baseline is the periodic ephemeris status report (256 B/cycle = 204.8 bps at 1 kbps), always present and topology-invariant. η_{cmd} uses $p_{\text{cmd}} = 1.0$ (command every active cycle) with $d = 0.10$ (10% of cycles active).

. *Step 3: Test B — TDMA Airtime (Algorithm 1, Lines 4–6)*

$$
\begin{aligned}
T_{\text{ing}} &= (k_c - 1) \times T_{\text{slot}} = 99 \times 80.0 = 7{,}920 \text{ ms} \\
T_{\text{egr}} &= T_{\text{cmd}} + T_{\text{hb}} + T_{\text{sync}} + T_{\text{turn}} \\
&= 160 + 24 + 4 + 4 = 192 \text{ ms} \\
T_{\text{ACK}} &= k_c \times 0.5 = 50 \text{ ms} \\
\text{margin} &= T_c - T_{\text{ing}} - T_{\text{egr}} - T_{\text{ACK}} \\
&= 10{,}000 - 7{,}920 - 192 - 50 = 1{,}838 \text{ ms} \quad \checkmark
\end{aligned}
$$

Annotation: Ingress is the dominant term (79.2% of T_c). Egress includes FEC-encoded command broadcast (512 B at 35 kbps with LDPC 7/8), heartbeat (64 B), re-sync preamble (120 bits), and two TX/RX turnarounds (2 × 2 ms). The 1,838 ms margin (18.4% of T_c) is the design target for accommodating ARQ and unmodeled overheads.

. *Step 4: ARQ Feasibility (Algorithm 1, Line 11)*

With $M_r = 1$ retransmission slot: $T_{\text{ARQ}} = 1 \times 80.0 = 80$ ms.

Residual margin: $1{,}838 - 80 = 1{,}758$ ms > 0 ✓.

Under GE losses: P95 demand = 14 failed nodes ×80 ms = 1,120 ms < 1,758 ms ✓.

Annotation: The ARQ demand is well within margin at 35 kbps. Compare with 30 kbps where margin is only 680 ms and P95 demand (1,288 ms) exceeds it—confirming the 35 kbps recommendation.

. *Step 5: Heuristic Cross-Check*

The design heuristic provides a lower-bound verification:

$$
\begin{aligned}
\alpha_{\text{RX}} &= T_{\text{ing}}/T_c = 7{,}920/10{,}000 = 0.792 \\
R_{\text{PHY,min}} &= \frac{C_{\text{coord,info}}}{\gamma \times \alpha_{\text{RX}}} = \frac{20.2}{0.732 \times 0.792} = 34.8 \text{ kbps}
\end{aligned}
$$

Since $34.8 < 35$ kbps ✓. The heuristic underestimates because it ignores egress time; it equals Test B exactly when $M_r = 0$ and egress = 0.

. *Summary*

All tests pass with comfortable margin. The 35 kbps PHY rate at $k_c = 100$ with $d = 0.10$ provides 18.4% superframe margin, accommodates $M_r = 1$ ARQ under GE losses (P95), and maintains Test A utilization at 30.2%. For other parameter combinations, substitute deployment-specific values into Algorithm 1 (core) or use the γ-conditional lookup in Table S4.